

Lecture 1

Application II: degeneracy, level splitting, and selection rules

Objectives. *Part I culminates in quantum mechanics. We make precise what it means for a group to be a symmetry of a Hamiltonian system — classically and quantum mechanically — and harvest the consequences: energy levels organise into symmetry-enforced multiplets whose sizes are dimensions of irreducible representations, while reduced-eigenvalue multiplicities and reality constraints determine total degeneracy; perturbations that lower the symmetry split the levels according to character restriction; and transition amplitudes obey selection rules governed by the Wigner–Eckart theorem, whose engine is tensor products and their Clebsch–Gordan decomposition. The infrared rule of Lecture 6 is derived, and invariant tensors open the bridge to Part II.*

1.1 Symmetry of a Hamiltonian system

Classical mechanics

Let a point transformation g act on configuration space, $\vec{x} \mapsto g\vec{x}$, with g orthogonal (the case we need). Its canonical lift to phase space takes momenta along, $(\vec{x}, \vec{p}) \mapsto (g\vec{x}, g\vec{p})$, and preserves Poisson brackets. The group G is a *symmetry of the Hamiltonian system* if the Hamiltonian is invariant,

$$H(g\vec{x}, g\vec{p}) = H(\vec{x}, \vec{p}) \quad \text{for all } g \in G.$$

The classical consequence is structural: since the transformation is canonical and preserves H , it maps solutions of Hamilton’s equations to solutions — if $t \mapsto (\vec{x}(t), \vec{p}(t))$ is a motion, so is its g -image. But note what a *finite* group does *not* give: a conserved quantity. Noether’s theorem trades a *continuous* family of symmetries for a constant of motion; a discrete symmetry like parity in a double-well potential conserves nothing, it merely pairs up trajectories. (The continuous story, and with it conservation laws, is the climax of Part II.)

Quantum mechanics

Quantum mechanics is kinder to discrete symmetries, for one reason: *states superpose*. The state space is a complex vector space, the symmetry acts on it linearly — by the function-space construction of Lecture 3,

$$(\rho(g)\psi)(\vec{x}) = \psi(g^{-1}\vec{x}),$$

unitary on square-integrable functions — and the whole apparatus of Part I applies. (That symmetries of quantum mechanics can always be implemented by unitary or antiunitary

operators is Wigner's theorem; for our point groups the unitary realisation above is all we need.)

Definition 1.1 (Quantum symmetry). A group G , unitarily represented by ρ on the state space, is a *symmetry of the quantum system* with Hamiltonian H if

$$[\rho(g), H] = 0 \quad \text{for all } g \in G.$$

Proposition 1.2. For $H = -\frac{1}{2m}\nabla^2 + V(\vec{x})$ and the function-space action above, $G \subset O(3)$ is a symmetry if and only if the potential is invariant: $V(g\vec{x}) = V(\vec{x})$ for all $g \in G$.

Proof. The substitution operator commutes with the Laplacian for any orthogonal g : by the chain rule, $\partial_\alpha[\psi(g^{-1}\vec{x})] = (g^{-1})_{\beta\alpha}(\partial_\beta\psi)(g^{-1}\vec{x})$, and summing the second derivatives,

$$\sum_\alpha (g^{-1})_{\beta\alpha} (g^{-1})_{\gamma\alpha} = (g^{-1}(g^{-1})^\top)_{\beta\gamma} = \delta_{\beta\gamma}$$

by orthogonality: $\nabla^2\rho(g) = \rho(g)\nabla^2$. The potential term acts by multiplication, and $\rho(g)V\rho(g)^{-1}$ is multiplication by $V(g^{-1}\vec{x})$; this equals multiplication by V for all g exactly when V is invariant (replace g by g^{-1}). ■

1.2 Degeneracy: the shape of a symmetric spectrum

Proposition 1.3. If $[\rho(g), H] = 0$ for all g , then every energy eigenspace $V_E = \{\psi : H\psi = E\psi\}$ is an invariant subspace: each eigenspace carries a representation of G .

Proof. $H\rho(g)\psi = \rho(g)H\psi = E\rho(g)\psi$. ■

How large are the eigenspaces, and which representations do they carry? The answer is the central theorem of Lecture 6 transplanted — the proof there used only $[D, \rho(g)] = 0$ and Schur's lemma, so it applies verbatim with the Hamiltonian in place of the dynamical matrix (restricted to any finite-dimensional invariant subspace, e.g. the span of finitely many levels; for unbounded H this is the mathematically honest formulation, and Proposition 1.3 is exact as stated):

In a symmetry-adapted basis the Hamiltonian is block-diagonal, with one reduced matrix $F^{(i)}$ of size $m_i \times m_i$ per irreducible representation, the same for every partner index. If a reduced eigenvalue has multiplicity q , its total complex degeneracy in that isotypic block is qn_i , where $n_i = \dim V^{(i)}$ is the symmetry-enforced multiplet size. If all reduced eigenvalues are simple and no unrelated blocks share an eigenvalue, each energy eigenspace is one irreducible multiplet.

Extra total degeneracy can arise in two algebraic ways: a repeated eigenvalue inside one reduced matrix (for example $F^{(i)} = \lambda\mathbb{I}_2$, giving $2n_i$) or a coincident eigenvalue from two unrelated reduced blocks. Both are accidental with respect to the stated complex symmetry and are generically lifted by symmetry-preserving perturbations. A third possibility is not accidental: in a real system, Lecture 6's conjugation constraint relates non-real conjugate irreducibles and forces their spectra to agree. Persistent degeneracy can therefore signal that reality/time reversal or a larger symmetry has not yet been included. The n^2 -fold levels of hydrogen — far exceeding the $(2\ell + 1)$ -fold rotational multiplets — are the canonical example, explained by a hidden four-dimensional rotational symmetry generated by the Runge–Lenz vector (Part II).

Example 1.4 (The square well: forced and accidental degeneracy side by side). A particle in a two-dimensional infinite square well, $(x, y) \in [-\frac{a}{2}, \frac{a}{2}]^2$, has the symmetry group D_4 of the square. The eigenfunctions factorise, $\psi_{nm} = \varphi_n(x)\varphi_m(y)$ with $E_{nm} \propto n^2 + m^2$, where φ_n is a cosine for odd n and a sine for even n , so that $\varphi_n(-x) = (-1)^{n+1}\varphi_n(x)$. Tracking the D_4 action on the pair $W_{nm} = \text{span}\{\psi_{nm}, \psi_{mn}\}$, $n \neq m$: the quarter-turn sends $\psi_{nm} \mapsto (-1)^{m+1}\psi_{mn}$ (off-diagonal), the half-turn multiplies by $(-1)^{n+m}$, the axis reflections multiply by $(-1)^{n+1}$, $(-1)^{m+1}$, and the diagonal reflection swaps the pair. The character of W_{nm} on the classes $(e, r^2, 2r, 2\sigma_v, 2\sigma_d)$ is therefore

$$\chi_W = (2, 2(-1)^{n+m}, 0, (-1)^{n+1} + (-1)^{m+1}, 0).$$

For $n + m$ odd this is $(2, -2, 0, 0, 0) = \chi_5$: the two-dimensional irreducible representation of D_4 (Lecture 5). The degeneracy of, say, E_{12} is *forced* by symmetry. For n, m of equal parity the character is $(2, 2, 0, \pm 2, 0)$, which decomposes into two *inequivalent one-dimensional* representations: the states $\psi_{nm} \pm \psi_{mn}$ belong to different symmetry types, and their shared energy — $E_{13} = E_{31}$, for instance — is an *accidental* degeneracy of the special separable potential, which a generic D_4 -symmetric perturbation of the well will split. The well even harbours purely arithmetic accidents, $E_{17} = E_{55}$ from $1 + 49 = 25 + 25$, connecting different (n, m) pairs altogether. One innocent system, all three phenomena.

1.3 Level splitting under symmetry reduction

Suppose a system with symmetry G is perturbed — an external field switched on, a ligand attached, an isotope substituted — so that only a subgroup $H \leq G$ survives as a symmetry. Each G -multiplet $V^{(i)}$ is still an H -representation by restriction, written $V^{(i)} \downarrow H$, but in general a *reducible* one; its H -irreducible content is read off by the standard character inner product, now taken over H :

$$V^{(i)} \downarrow H \cong \bigoplus_j b_{ij} W^{(j)}, \quad b_{ij} = \left\langle \chi_j^H, \chi_i|_H \right\rangle_H,$$

the *branching multiplicities*. The perturbed Hamiltonian commutes with the H -action, so by the block-diagonalisation theorem the formerly degenerate level decomposes into H -irreducible constituents. Each $W^{(j)}$ supplies an enforced multiplet of size $\dim W^{(j)}$; equivalent repeated constituents may mix through a reduced matrix, and conjugate constituents may remain paired by a real/antiunitary constraint. Character arithmetic gives the allowed splitting pattern, while reduced eigenvalues, their multiplicities, magnitudes, and ordering require dynamics.

Example 1.5 ($C_{3v} \rightarrow C_s$: splitting a doublet). Keep one mirror plane of ammonia and discard the rest: $H = C_s = \{E, \sigma\}$, with characters $A' = (1, 1)$ and $A'' = (1, -1)$. Restricting the C_{3v} table (Lecture 5) to these two elements: $A_1 \rightarrow A'$, $A_2 \rightarrow A''$, and for the doublet, $\chi_E|_{C_s} = (2, 0) = A' \oplus A''$. A twofold degenerate E level splits into two nondegenerate levels, one symmetric and one antisymmetric under the surviving mirror. Physically: replace one hydrogen of NH_3 by deuterium. The *force field* (the potential) retains full C_{3v} symmetry — electromagnetism cannot tell H from D — but the masses enter the dynamical matrix through the mass-weighting of Lecture 6, and the mass distribution has only C_s symmetry: in NH_2D each doubly degenerate E vibration of ammonia splits into a close pair of lines, exactly as observed.

Example 1.6 (Crystal-field splitting of d levels). A free atom is rotationally symmetric, and Part II will show that its $(2\ell+1)$ -fold multiplets are irreducible representations of the rotation group with characters

$$\chi^{(\ell)}(C(\theta)) = \frac{\sin(\ell + \frac{1}{2})\theta}{\sin \frac{\theta}{2}} \quad (1.1)$$

on a rotation through θ — a formula we borrow now and repay with proof in Part II. Place the atom in an octahedral crystal environment (six identical ligands on the axes): the surviving rotation group is the octahedral group $O \cong S_4$ of Lecture 2, with classes E , $8C_3$, $3C_2$, $6C_4$, $6C'_2$ and character table

O	$1 E$	$8 C_3$	$3 C_2$	$6 C_4$	$6 C'_2$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
E	2	-1	2	0	0
T_1	3	0	-1	1	-1
T_2	3	0	-1	-1	1

(the full point group $O_h = O \times \{E, i\}$ adds only a parity label, even for d orbitals, by Lecture 2's direct-product structure). For a d level, $\ell = 2$, formula (1.1) at $\theta = 2\pi/3, \pi, \pi/2, \pi$ gives

$$\chi_d = (5, -1, 1, -1, 1),$$

and the inner products over $|O| = 24$ yield

$$m_{A_1} = m_{A_2} = m_{T_1} = 0, \quad m_E = \frac{1}{24}(10 + 8 + 6) = 1, \quad m_{T_2} = \frac{1}{24}(15 - 3 + 6 + 6) = 1 :$$

$$d \longrightarrow E \oplus T_2.$$

The fivefold d level splits into a doublet (e_g , in the chemists' O_h labels) and a triplet (t_{2g}) — the celebrated crystal-field splitting whose magnitude sets the absorption bands, and hence the colours, of transition-metal complexes. The same computation runs unchanged for p and f levels and for tetrahedral coordination — a worthwhile exercise.

1.4 Tensor products and Clebsch–Gordan decomposition

Composite objects in quantum mechanics — two subsystems, a state and an operator, an orbital and a vibration — live in tensor products, and the group acts on all factors at once.

Definition 1.7 (Tensor product of representations). Given vector spaces V, W with bases $(e_\mu), (f_\nu)$, the *tensor product* $V \otimes W$ is the vector space with basis the nm symbols $e_\mu \otimes f_\nu$, and $v \otimes w$ is defined for general vectors by bilinear expansion. (The construction is independent of the chosen bases up to canonical isomorphism; we will not need the basis-free formulation.) Given representations (ρ_V, V) and (ρ_W, W) of the same group G , the *tensor product representation* acts factorwise:

$$(\rho_V \otimes \rho_W)(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w,$$

extended linearly; the homomorphism property is immediate factor by factor.

Proposition 1.8 (Characters multiply). $\chi_{V \otimes W} = \chi_V \cdot \chi_W$.

Proof. In the product basis the matrix is $D_{\mu\alpha,\nu\beta} = D_{\mu\nu}^V D_{\alpha\beta}^W$, with trace $\sum_{\mu,\alpha} D_{\mu\mu}^V D_{\alpha\alpha}^W = \chi_V \chi_W$. ■

Products of irreducibles are generally reducible, and characters decompose them instantly:

Definition 1.9 (Clebsch–Gordan series and coefficients). The decomposition

$$V^{(i)} \otimes V^{(j)} \cong \bigoplus_k c_{ij}^k V^{(k)}, \quad c_{ij}^k = \langle \chi_k, \chi_i \chi_j \rangle_G,$$

is the *Clebsch–Gordan series*. The entries of a unitary change of basis from the product basis $e_\mu^{(i)} \otimes e_\nu^{(j)}$ to a symmetry-adapted basis of the right-hand side are the *Clebsch–Gordan coefficients* $\langle k, \kappa, (s) | i, \mu; j, \nu \rangle$, with s indexing the c_{ij}^k copies when the multiplicity exceeds one.

Example 1.10 (Clebsch–Gordan table of C_{3v}). With the character table of Lecture 5: A_1 is the identity of the product, $A_2 \otimes A_2 = A_1$, and $A_2 \otimes E$ has character $(1, 1, -1) \cdot (2, -1, 0) = (2, -1, 0)$, so $A_2 \otimes E = E$. The interesting entry is $\chi_E^2 = (4, 1, 0)$:

$$m_{A_1} = \frac{1}{6}(4 + 2) = 1, \quad m_{A_2} = \frac{1}{6}(4 + 2) = 1, \quad m_E = \frac{1}{6}(8 - 2) = 1,$$

$$E \otimes E = A_1 \oplus A_2 \oplus E.$$

Two refinements of the character calculus will be used repeatedly.

Proposition 1.11 (Symmetric and antisymmetric squares). $V \otimes V$ splits invariantly into symmetric and antisymmetric parts, $S^2V \oplus \Lambda^2V$, with characters

$$\chi_{S^2V}(g) = \frac{1}{2}[\chi(g)^2 + \chi(g^2)], \quad \chi_{\Lambda^2V}(g) = \frac{1}{2}[\chi(g)^2 - \chi(g^2)].$$

Proof. The symmetrised products $e_\mu \otimes e_\nu \pm e_\nu \otimes e_\mu$ span invariant subspaces, since the factorwise action respects the swap. For the characters, diagonalise $\rho(g)$ (possible: finite order) with eigenvalues λ_μ ; the symmetric square has eigenbasis indexed by $\mu \leq \nu$ with eigenvalues $\lambda_\mu \lambda_\nu$, and

$$\sum_{\mu \leq \nu} \lambda_\mu \lambda_\nu = \frac{1}{2} \left[\left(\sum_\mu \lambda_\mu \right)^2 + \sum_\mu \lambda_\mu^2 \right] = \frac{1}{2} [\chi(g)^2 + \chi(g^2)],$$

since $\sum \lambda_\mu^2 = \text{tr } \rho(g)^2 = \chi(g^2)$. Subtract for the antisymmetric part. ■

(The symmetric square is the home of the quadratics $x^2, xy, \dots$ — the Raman operators of Lecture 6; the antisymmetric square will return in Part II as the origin of the cross product and of the adjoint representation.)

Proposition 1.12 (Pairing with the dual). For irreducible $V^{(i)}, V^{(j)}$, the trivial representation occurs in $V^{(i)} \otimes V^{(j)}$ if and only if $V^{(j)} \cong (V^{(i)})^*$, and then exactly once.

Proof. The multiplicity of the trivial representation is

$$\langle \chi_{\text{triv}}, \chi_i \chi_j \rangle_G = \frac{1}{|G|} \sum_g \chi_i(g) \chi_j(g) = \langle \bar{\chi}_i, \chi_j \rangle_G = \langle \chi_{i^*}, \chi_j \rangle_G = \delta_{i^*, j},$$

using $\chi_{i^*} = \bar{\chi}_i$ (Lecture 5) and the first orthogonality relation. ■

1.4.1 Invariant tensors: the bridge to Part II

A vector T in a representation with $\rho(g)T = T$ for all g — equivalently, a copy of the trivial representation — is an *invariant*; an invariant in a tensor power $V \otimes \cdots \otimes V$ (possibly with dual factors) is an *invariant tensor*. Their count is one character integral,

$$\dim\{\text{invariants}\} = \langle \chi_{\text{triv}}, \chi \rangle_G = \frac{1}{|G|} \sum_g \chi(g),$$

the multiplicity of the trivial representation. For the standard representation of S_3 , with $\chi = (2, 0, -1)$: in $V \otimes V$, $\chi^2 = (4, 0, 1)$ gives $\frac{1}{6}(4+0+2) = 1$ invariant — the invariant bilinear form, i.e. the $\delta_{\mu\nu}$ left untouched by the orthogonal matrices; and in $V \otimes V \otimes V$, $\chi^3 = (8, 0, -1)$ gives $\frac{1}{6}(8+0-2) = 1$ invariant cubic — a symmetric three-index tensor $d_{\mu\nu\lambda}$ peculiar to the triangle's geometry. Invariant tensors are the rigid skeletons of a symmetry: every invariant interaction, coupling, or form a theory can contain must be built from them. In Part II this principle, applied to $SU(2)$, will dictate which terms may appear in physical Hamiltonians and Lagrangians; scaled up to $SU(3)$ in explicitly further study, the same idea fixes the δ , ϵ and d , f tensors used in strong-interaction physics. That $SU(3)$ application, including its triplet/antitriplet and eightfold-way content, is not taught in this course; Jones, Ch. 8, §§8.3–8.4 is the reading destination.

1.5 The Wigner–Eckart theorem and selection rules

Operators, like states, organise into symmetry multiplets. The group acts on operators by conjugation, $A \mapsto \rho(g)A\rho(g)^{-1}$ — a representation on the space of operators, equivalent to $V \otimes V^*$, with character $|\chi_V|^2$ — and Maschke applies to it like to any other.

Definition 1.13 (Irreducible tensor operator). A family of operators $T_\nu^{(j)}$, $\nu = 1, \dots, n_j$, is an *irreducible tensor operator of type j* if it transforms among itself with the matrices of $V^{(j)}$:

$$\rho(g) T_\nu^{(j)} \rho(g)^{-1} = \sum_\mu D_{\mu\nu}^{(j)}(g) T_\mu^{(j)} \quad \text{for all } g \in G.$$

Type $j = \text{trivial}$ means $[\rho(g), T] = 0$: an *invariant* (scalar) operator, such as the Hamiltonian itself. The components of the electric dipole moment transform like the coordinates (x, y, z) and furnish tensor operators of the corresponding vector types.

Theorem 1.14 (Wigner–Eckart, finite groups). Let $|i, \mu, a\rangle$ be symmetry-adapted states (copy a of type i , partner μ) and $T_\nu^{(j)}$ an irreducible tensor operator. Then

$$\langle k, \kappa, b | T_\nu^{(j)} | i, \mu, a \rangle = \sum_{s=1}^{c_{ji}^k} \langle k, \kappa, (s) | j, \nu; i, \mu \rangle \langle k, b || T^{(j)} || i, a \rangle_s :$$

the dependence on all three partner indices κ, ν, μ sits entirely in Clebsch–Gordan coefficients — pure group theory — while the dynamics enters only through c_{ji}^k reduced matrix elements per triple of types. In the common multiplicity-free case $c_{ji}^k \leq 1$ there is a single product of one CG coefficient and one reduced matrix element; and if $c_{ji}^k = 0$ the matrix element vanishes identically.

Proof. Define $\Phi: V^{(j)} \otimes V^{(i)} \rightarrow \mathcal{H}$ on the basis by $\Phi(f_\nu \otimes e_\mu) = T_\nu^{(j)} |i, \mu, a\rangle$. This map intertwines: using the definition of a tensor operator and the adapted transformation of the

states,

$$\rho(g) T_{\nu}^{(j)} |i, \mu, a\rangle = (\rho(g) T_{\nu}^{(j)} \rho(g)^{-1}) \rho(g) |i, \mu, a\rangle = \sum_{\nu', \mu'} D_{\nu' \nu}^{(j)}(g) D_{\mu' \mu}^{(i)}(g) T_{\nu'}^{(j)} |i, \mu', a\rangle,$$

which is Φ applied to the product action on $f_{\nu} \otimes e_{\mu}$. Now compose Φ with the equivariant projection onto the b -th type- k copy in $\mathcal{H}$: the result is an intertwiner $V^{(j)} \otimes V^{(i)} \rightarrow V^{(k)}$. By the Clebsch–Gordan series and the multiplicity lemma of Lecture 4, the space of such intertwiners has dimension exactly c_{ji}^k — zero, and the matrix element vanishes, or spanned by the c_{ji}^k canonical intertwiners whose matrix entries are the Clebsch–Gordan coefficients. Expanding the composed map in that basis, with coefficients christened the reduced matrix elements, and taking the κ -component, gives the formula. ■

The theorem industrialises the manoeuvre that proved the central theorem of Lecture 6 (where T was the invariant operator D and the “reduced matrix elements” were the entries of $F^{(i)}$). Its sharpest edge is the vanishing statement:

Corollary 1.15 (Selection rules). *A transition amplitude $\langle f | T | i \rangle$, with the final states, the operator, and the initial states transforming by Γ_f , Γ_T , Γ_i , vanishes identically unless Γ_f occurs in $\Gamma_T \otimes \Gamma_i$ — equivalently (Proposition 1.12), unless the trivial representation occurs in*

$$\Gamma_f^* \otimes \Gamma_T \otimes \Gamma_i.$$

Example 1.16 (Infrared activity, derived at last). For a molecule in its vibrational ground state $|0\rangle$ — a product of Gaussians in all normal coordinates, invariant under the point group, hence type A_1 — the absorption of one infrared photon into mode family i has amplitude $\langle 1_i, \mu | d_{\alpha} | 0 \rangle$, where the dipole components d_{α} transform as Γ_{vec} , the representation of (x, y, z) . One-quantum states transform like their modes: $|1_i, \mu\rangle = a_{i\mu}^{\dagger} |0\rangle$ with the creation operator linear in the mode coordinate. The selection rule demands the trivial representation inside $V^{(i)*} \otimes \Gamma_{\text{vec}} \otimes A_1$, i.e. (point-group characters here being real, $V^{(i)*} \cong V^{(i)}$) that $V^{(i)}$ occur in Γ_{vec} — precisely the rule stated in Lecture 6, now a theorem. For Raman scattering the operator is the polarisability, transforming as the quadratics inside $S^2 \Gamma_{\text{vec}}$ (Proposition 1.11), and the rule reads the same with quadratics in place of coordinates. In a centrosymmetric molecule the inversion i acts as -1 on all coordinates, hence as -1 on Γ_{vec} (odd, u) and as $+1$ on every quadratic (even, g); a mode of definite parity can match one list or the other but never both: *mutual exclusion*, in two lines, as promised.

1.6 Part I in retrospect

Seven lectures ago, symmetry was a collection of operations on a triangle. The arc since: operations abstracted to groups (Lecture 1) and their internal anatomy — conjugacy, quotients, products — mapped out (Lecture 2); groups set acting on linear spaces, with the function-space construction $\psi \mapsto \psi \circ g^{-1}$ as the universal mechanism, and complete reducibility guaranteed (Lecture 3); Schur’s lemma converting symmetry into rigid constraints, counted by $\sum n_i^2 = |G|$ and the class theorem (Lecture 4); characters compressing it all into tables, with projectors to make it operational (Lecture 5); and the payoff in physics: vibrational spectra shaped by representation theory (Lecture 6), and now energy levels, their splittings, and the very visibility of transitions governed by the same machinery (Lecture 7). Throughout, one engine did the heavy lifting — *average over the group, and invoke Schur* —

and one slogan summarises the physics: *what symmetry permits is organised by irreducible representations; what symmetry forbids is exactly zero.*

Part II asks what survives when the group becomes continuous: rotations through every angle, not just multiples of $2\pi/n$. Sums over the group become normalised Haar integrals. For finite-dimensional continuous representations of compact groups, averaging again gives an invariant inner product and complete reducibility; Schur's lemma remains linear algebra; and the same averaging argument gives matrix-element and character orthogonality and integral projector formulae. Lecture 12 proves these statements for $U(1)$ and $SU(2)$ and develops the corresponding Wigner–Eckart theorem. The finite counting identities $\sum_i n_i^2 = |G|$ and “number of irreducibles equals number of conjugacy classes”, and the finite group-algebra decomposition behind them, do *not* survive literally. Their compact-group replacement is Peter–Weyl: irreducible matrix elements are dense in $L^2(G)$, and irreducible characters are complete among class functions. But continuity adds what no finite group possesses: the ability to differentiate. Infinitesimal symmetries form a new algebraic structure — a Lie algebra — whose representation theory yields angular momentum, spin, and the spherical harmonics (repaying the character formula (1.1) borrowed above); Noether's theorem finally converts symmetry into conservation laws; and invariant tensors — developed for $SU(2)$ here, while $SU(3)$ is explicitly further study — prepare the vocabulary for particle-physics applications. The triangle has carried us as far as it can; from Lecture 8, the circle takes over.

Remark 1.17 (Computer algebra). The notebook for this lecture automates the Clebsch–Gordan series for any point group — products of character rows, decomposed by the inner product — reproducing the C_{3v} table of Example 1.10. The Wigner–Eckart theorem is then seen at work in the square well of Example 1.4: the dipole matrix elements $\int \overline{\psi_{n'm'}} x \psi_{nm}$, computed as elementary integrals, vanish exactly when Corollary 1.15 says they must, and the nonvanishing ones between members of degenerate doublets exhibit the ratios fixed by the D_4 Clebsch–Gordan coefficients — group theory predicting integrals before they are computed.

Summary.

- *Degeneracy.* Each energy eigenspace of a G -invariant Hamiltonian carries a representation of G . On a complex isotypic block, a reduced eigenvalue of multiplicity q has total degeneracy $q \dim V^{(i)}$; $\dim V^{(i)}$ is the enforced multiplet size, with conjugate sectors paired when reality requires it (Proposition 1.3; Lecture 6).
- *Level splitting.* A perturbation invariant only under a subgroup $H \leq G$ splits a multiplet according to the restriction to H of its character.
- *Clebsch–Gordan.* $\chi_{V \otimes W} = \chi_V \chi_W$; reducing the product character decomposes $V \otimes W$ into irreducibles (Proposition 1.8).
- *Wigner–Eckart.* Matrix elements of a tensor operator factorise into a Clebsch–Gordan coefficient and one reduced matrix element (Theorem 1.14), whence the selection rules (Corollary 1.15).
- *$SO(3)$ characters.* $\chi^{(\ell)}(\theta) = \sin((\ell + \frac{1}{2})\theta)/\sin(\theta/2)$ (eq. (1.1); established in Lecture 12).

Tutorial. Exercise Sheet 7.

References

Tinkham, *Group Theory and Quantum Mechanics* (selection rules and the Wigner–Eckart theorem); Elliott–Dawber, *Symmetry in Physics*; Tung, *Group Theory in Physics*, Ch. 4.

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: 10.1016/0375-9601(75)90798-7.
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: 10.1103/PhysRevLett.35.1053.
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.